

AccessGuard++: AI-Enhanced Risk Intelligence for Zero Trust Systems

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Abstract— The cyberattacks are increasingly advanced, conventional perimeter security models are not longer sufficient to cover key digital infrastructures. This paper proposes AccessGuard, a real-time, risk-aware Zero Trust security framework that combines behavioural anomaly detection based on machine learning. Unlike static authentication techniques, AccessGuard continuously evaluates contextual risk factors such as device posture, location, and access frequency while modelling user and device behaviour to identify anomalies. According to real-time risk scoring, The system employs adaptive access control policies, which include step-up authentication or denial of access in high risk conditions to non-problematic access in low risk conditions. AccessGuard is an active and flexible business security platform to enhance resilience to insider threats, misuse of credentials, and advanced persistent attacks through a combination of intelligent anomaly detection and the implementation of Zero Trust principles in a contemporary business.

Keywords— Zero Trust security, Risk-aware access control, Behavioural anomaly detection, Contextual risk assessment, Credential misuse prevention

I. INTRODUCTION

The digital transformation of the industries (distributed systems, cloud-based systems, and remote access) has been very rapid and has transformed the landscape of IT. Although the technologies enhance efficiency and scalability, they also leave a larger and more susceptible attack surface. Consequently, the old-fashioned perimeter-based security frameworks where trust is placed on anything within the network edge are not effective anymore in the present times against contemporary threats such as APTs, credential theft, and insider attacks. In order to address these constraints, companies need to transition to a new system the Zero Trust Security Model (ZTM).

[1] The philosophy of ZTM is never trust, always verify, and it recognizes the fact that the internal and external environment can be at risk. [2] ZTA is closely connected with Identity and Access Management (IAM) and provides a dynamic security model to ensure that every access request is authenticated and authorized by mandate, whether an entity is located on the private side or the public side, and whether it has the history of accessing the system or not. Nevertheless,

most of the existing Zero Trust deployments tend to rely on fixed rules or predefined policies that are configured on basic contextual cues. This dependence on constant parameters introduces a serious weakness as these systems are unable to identify complex and low and slow attacks whereby bad actors carefully replicate the normal user behavior to avoid detection.

To solve this security gap, this paper introduces the proposed solution AccessGuard, a zero-trust architecture, which is real-time and risk-aware. AccessGuard is equipped with state-of-the-art behavioral anomaly detection that runs on machine learning. It is a continuous assessment of risk whereby it keeps contextualizing attributes like geolocation, device posture, access time, and past historical user/device behavior. The continuous assessment will allow identifying anomalies in real time and preventing unauthorized access attempts before they cause damage.

The system uses adaptive access control policies, which are calculated on the basis of the risk score. A low-risk scenario results in users being able to access the system with ease; a medium-risk scenario results in step-up authentication, e.g. biometric verification; and a high-risk scenario, access is denied altogether. This renders AccessGuard smart, adaptable and much more efficient as compared to the conventional fixed security systems. AccessGuard provides a unique and innovative way of enhancing organizational resilience to credential abuse and insider threats by integrating the fundamental principles of Zero Trust with the latest developments in anomaly detection to protect the current digital infrastructure.

A. Problem Statement

Although a wide range of advanced security models have been embraced, organizations around the world are still grappling with the problem of detecting and preventing complex attacks like Advanced Persistent Threats (APTs), insider attacks, and credential abuse. When attackers gain legitimate credentials, traditional perimeter-based security models do not work anymore because they have implicit faith in the parties located on the other side of the network. Although the Zero Trust [3]

Even though ZTM is based on the continuous verification, the majority of the current version of implementations are

based on the fixed, pre-determined rules, which poses a significant security gap.

Such traditional approaches do not have the dynamism to respond to user behavior in real-time and cannot usually identify more complex, low-and-slow attacks that carefully simulate legitimate user actions. Moreover, most of the current Zero Trust architectures do not include the necessary functionality of real-time risk identification and cognitive anomaly detection to create not the least critical blind spots on which sensitive digital infrastructures are susceptible to unauthorized access and data breaches [3]. But the bottom line requirement is to come up with a risk-sensitive, intelligent system that transcends a fixed policy implementation. In particular, there is an urgent need of a solution which would combine [4] machine-learning-based anomaly detection with Zero Trust principles And Artificial Intelligence (AI), particularly Machine Learning (ML), offers immense potential for augmenting ZTA by automating threat detection and enabling real-time response strategies to offer dynamic, proactive, and context-aware access control. This need is suggested to be satisfied with the help of AccessGuard that concentrates on continuous and behavior-based risk assessment.

II. LITERATURE SURVEY

[2] As it has been mentioned in this paper, Zero Trust Architecture (ZTA) is a good alternative to perimeter-based security because it introduces the concept of no trust, no trust, which should remain its motto. The Zero Trust Architecture (ZTA) is aimed at minimizing the attack surfaces through constant authentication, granting access only to the bare minimum of the resources and breaking down the networks into smaller, secure units (Rose et al., 2019; Hasan, 2024). According to the latest research, these systems could be further enhanced with artificial intelligence and machine learning so that they are more efficient in detecting abnormal behaviors and access control in enterprise networks, cloud systems, or future 6G technologies (Ganesh et al., 2024). There is a common agreement among the literature that ZTA is becoming an essential standard in modern cybersecurity strategies even though there are still widespread cost and scalability concerns.

[5] This paper examines ways that Microsoft Copilot can be used to support cybersecurity operations in combining with Zero Trust Architecture (ZTA). It highlights how AI-enabled through the use of anomaly monitoring and natural conversations, solutions may enhance threat detection, automate Identity and Access Management (IAM) and allow least privilege access (Rana, 2025). The practical examples provided, including Intesa Sanpaolo Bank, demonstrate that Copilot simplifies the process of incident response and can help the new analyst to be more productive sooner. The report mentions such positive aspects as scaling, continuous testing, and cost minimization along with risks such as overdependence on AI, data privacy issues, and adversarial attacks. On balance, studies have shown that AI and ZTA integration can assist organizations to develop a stronger and more flexible cybersecurity posture.

This paper discusses the implementation of Microsoft Copilot with a Zero Trust Architecture (ZTA) to optimize the cybersecurity operation. It illustrates that AI solutions can assist in enhancing threat detection with the help of anomaly monitoring and natural language interactions, automate Identity and Access Management (IAM) and allow least

privilege access (Rana, 2025). Copilot has demonstrated itself in a real-world case, like Intesa Sanpaolo Bank, to decrease the time of incident response and make junior analysts more productive in less time.[6] The paper describes such advantages as scalability, real-time monitoring, and cost-reduction, and also outlines threats such as the over-reliance on AI, data privacy, and possible adversarial attacks. In general, the literature suggests that AI and ZTA can be used to assist organizations to achieve a more resilient and versatile cybersecurity posture.

III. METHODOLOGY

AccessGuard is anonymously proposed system that deploys risk-aware, real-time Zero Trust security model, which is a combination of adaptive access control and continuous monitoring of behavior. The process shown in the process diagram can be split into three major stages: Registration and Baseline Establishment, Real-Time Risk Assessment, and Adaptive Access Decision.

Registration and Baseline Establishment
When a user is first registered, registration process is performed whereby a username, password, as well as a live video snapshot is obtained. The result of this process is a Baseline Trust Profile that comprises original security and behavioral metadata (registered IP, device, and location). Such profile is a benchmark in the determination of the credibility of future sessions.

A. Real-Time Risk assessment

Every time a user logs or enters a session, the database gathers real-time contextual data (e.g. device, geolocation, access time, and use patterns). [7] A Feature Extraction and Trust File Comparison process is done on these session attributes. The current behavior that would be considered a significant deviation with reference to the Baseline Trust Profile is identified as a possible anomaly. The extracted features are fed to the Machine Learning Risk Engine (as implemented in risk.py) which has been pre-trained on previous data of past sessions. The engine provides a dynamic risk score (scaled between 0 to 100) on the session depending on the similarity to the acceptable historically observed behavior. A low score means that there is normal activity and a high score is that there is suspicious or unusual behavior. Any security alerts, flagged devices, and risk scores are logged and viewed on the Admin Risk Dashboard to monitor it continuously.

B. Adaptive Access Decision Engine

[8] The Access Decision Engine considers the corresponding risk score and implements the corresponding adaptive security measure: Path 1: Low Risk (Score: 0), - Direct Allow. When the risk score falls within the range in the range of [0,30], the system considers it as a low risk and permits it directly to access the system (Result Allow). Path 2: Medium Risk (Score: 31), - Step-Up MFA. When the risk score falls in the range [31,59] This is followed by a Real Time Face Detection check. In case the face of the user is recognized and matched (Result Allow), the access is permitted (Result Allow); in case the score is not high (above 70) (Result Block), the entry is blocked. Path 3: High Risk (Score>0.70) - Immediate Block. Any session whose score exceeds 70 is considered to be a high-risk. Here, instant blockage of access is produced (Result Block) and an Admin Risk Dashboard Real Time Alert is produced.

The system also keeps on updating the learning model with current session data. This allows AccessGuard to dynamically adjust to dynamic user behaviour and changing security risks to provide greater protection over time, through behavioural anomaly detection in combination with dynamic access control, providing a risk-aware, real-time Zero Trust framework that is more resistant to insider threats, advanced persistent attacks, and credential abuse.[3]The solution is consistent with previous research suggesting intelligent infrastructure based on the Zero Trust concepts of remote infrastructure infrastructure, including the ZT-VPN architecture.

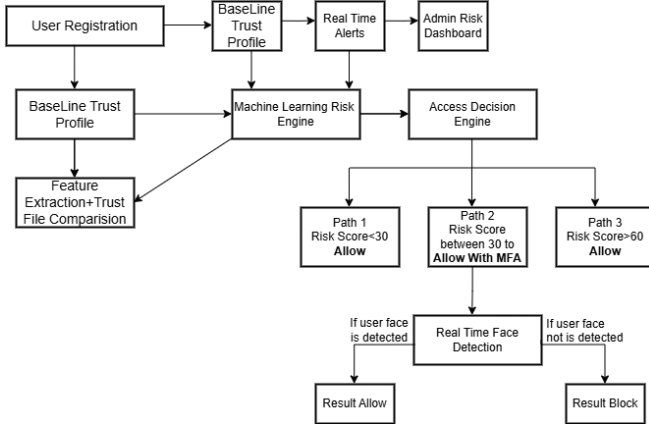


Fig. 1. Block Diagram

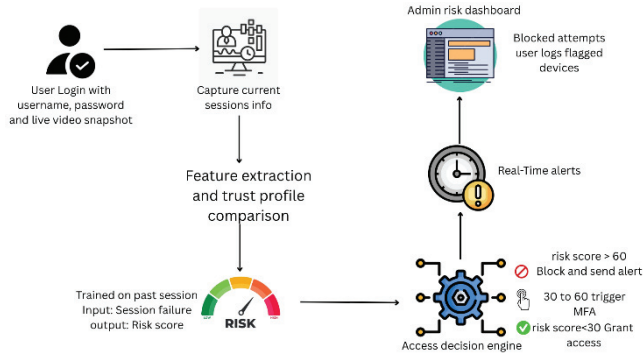


Fig. 2. System Architecture

The system architecture [Fig. 2] demonstrates the work flow of the project

Mathematically Description:

$S = \{I, O, F, DD, NDD, \text{Failure}, \text{Success}\}$

Where,

S =System

I = Input

O =Output

F =Failure

DD = Deterministic Data

NDD = Non-Deterministic Data

Failure = Failure conditions

Success = Success conditions

I is Input of system

Input I = set of Inputs

Where,

$I = \{I1, I2, I3, I4\}$

Where,

$I1 = \{\text{Admin}\}$

$I2 = \{\text{HR}\}$

$I3 = \{\text{Devices}\}$

$I4 = \{\text{Resources \& Telemetry}\}$

F is Function of system

F = set of Function

Where,

$F1 = \{\text{Register / Enroll User}\}$

$F2 = \{\text{Authenticate (credentials, SSO)}\}$

$F3 = \{\text{Collect Session \& Device Telemetry}\}$

$F4 = \{\text{Feature Extraction (contextual \& behavioural)}\}$

$F5 = \{\text{Baseline Trust Profile Creation / Update}\}$

$F6 = \{\text{ML-based Anomaly Detection \& Risk Scoring}\}$

$F7 = \{\text{Access Decision Engine (map risk} \rightarrow \text{action)}\}$

$F8 = \{\text{Step-up Auth (MFA) / Enforcement}\}$

$F9 = \{\text{Alerting, Logging \& Admin Dashboard}\}$

$F10 = \{\text{Continuous Learning / Model Retraining}\}$

O is Output of system

Output

$O1 = \{\text{Access Granted}\}$

$O2 = \{\text{MFA (Multi-Factor Authentication) Required}\}$

$O3 = \{\text{Access Blocked \& Security Alert Triggered}\}$

$O4 = \{\text{Admin Reports \& Audit Logs Generated}\}$

Success Conditions: When the system correctly identifies anomalies, implements zero-trust policies, and makes sure that authorised users are granted access while malicious attempts are prevented, it is considered successful. As a result, protection is safe, dependable, and instantaneous.

Failure Conditions: Failure happens when the system misclassifies user behaviour, when behavioural data is incomplete or tainted, or when technical problems like network outages hinder real-time enforcement.

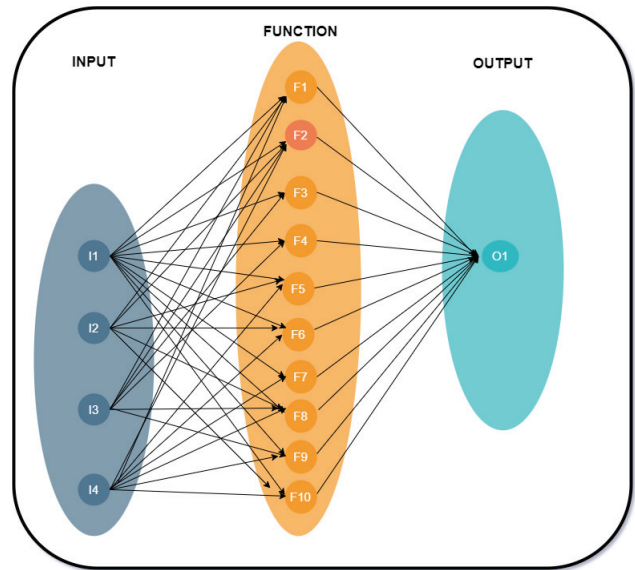


Fig. 3. Functional Dependency Diagram

TABLE I. FUNCTIONAL DEPENDENCY TABLE

	F 1	F 2	F 3	F 4	F 5	F 6	F 7	F 8	F 9	F 10
F 1	1	0	0	0	0	0	0	0	0	0
F 2	0	1	0	0	0	0	0	0	0	0
F 3	0	0	1	0	0	0	0	0	0	0
F 4	0	0	0	1	0	0	0	0	0	0
F 5	0	0	0	0	1	0	0	0	0	0
F 6	0	0	0	0	0	1	0	0	0	0
F 7	0	0	0	0	0	0	1	0	0	0
F 8	0	0	0	0	0	0	0	1	0	0
F 9	0	0	0	0	0	0	0	0	1	0
F 10	0	0	0	0	0	0	0	0	0	1

IV. DEPLOYMENT

The Flow diagram (Fig. 4) shows the configuration of the run-time processing nodes and the many parts that are housed within them. Users, including administrators, employers, and students, access the system using client devices, such as PCs, laptops, or mobile devices. Incoming requests from these clients are routed to the web servers, which efficiently distribute traffic and deliver the necessary web pages. Together with the web servers, the application server manages user requests, manages business logic, and enables smooth communication with respect to system components like employer administration, training analysis, and placement management.

All crucial data on students, companies, placements, and training is safely stored and managed by the database server, which also ensures the data's dependability and accuracy. [9]

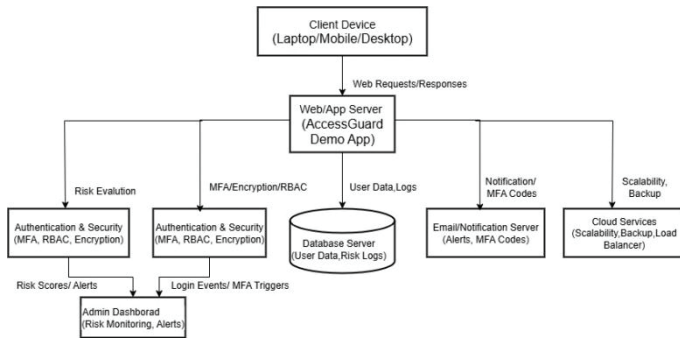


Fig. 4. Flow Diagram

Functionality:

1. Provides a secure and reliable platform for students and employers to participate in campus recruitment and placement activities.
2. Offers data-driven analysis for training programs and recruitment processes.
3. Ensures seamless communication and interaction through email services.
4. Integrates with university systems to leverage existing data and resources effectively.

The deployment diagram illustrates how these components interact to deliver a robust solution for integrated campus recruitment, placement, and training analysis.

V. PROPOSED MODULE

AccessGuard is designed as a modular framework, where each component works in coordination to deliver risk-aware, real-time Zero Trust security powered by behavioural anomaly detection.

A. Identity and Access Management (IAM) Module:

The security component that responds to the identification and authorization of the users is Identity and Access Management (IAM) Module; it is the main component of AccessGuard. [10] IAM plays a critical role in securing digital assets and ensuring that only authorized users can access sensitive systems and data. It has Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC). MFA provides an added layer of protection by requiring users to provide multiple forms of verification. RBAC allows organizations to assign permissions based on users' roles, ensuring that employees only have access to the data necessary for their tasks.[10] The IAM identifies users who may access the system through authentication of the user identity credentials and uses contextual access policies.

B. Device Condition and Context Assessment Module:

The device is judged in terms of its security posture before it is allowed access to the system.[11] Understand the health and status of devices to inform risk decisions. Real time inspection, assessment and patching informs every access request. The parameters which are going to be considered are the version of operating system, patch status, security health and indicators of compromise. Moreover, this module gathers real time contextual information, such as the connecting IP address, geolocation and time of login and is fed into the subsequent risk analyzer.

C. Risk Scoring and Policy Engine (RSP Engine):

This is the main computational core. The RSP Engine computes risk scores on a per-use basis by summing various identity verifications, machine-posture data as well as machine learning based behavior analysis. The RSP Engine is designed to always predict user and device behavior so as to identify anomalies that are statistically significant.[12] This framework proposes multi-layered intelligence that incorporates behavioural analytics, anomaly detection, and dynamic policy orchestration, in which continuous verification and risk-based access decisions are enabled. This paper proposes and investigates a model of Dynamic Risk-Adaptive Access Control (DRAAC), leveraging machine learning techniques to continuously assess user and device risk scores, enforce fine-grained policies, and reduce insider and outsider threats.[13] The Policy Engine then makes a decision based on this real-time risk score, encompassing seamless access control (a low-risk (score <30)) or step-up authentication (e.g., face detection) for a medium-risk (score 30-59)) or total denial of access (high-risk (score >60)) decision.

D. Monitoring and Auditing Module :

This is a mechanism that facilitates constant monitoring of the activities of the system through close accounting of all access requests, risk evaluation and anomaly detection. It provides administrators with the capability to generate reports of compliance and investigate past trends as a way of

enhancing security. This is further enabled by the introduction of ML-based anomaly detection techniques that allow the system to identify small and long run insider threats that may otherwise not be transparent with normal monitoring systems.[14]The incorporation of Machine Learning specifically enables Automated Threat Detection, Real-Time Risk Assessment, and Regulatory Compliance Auditing within cybersecurity risk management models.

E. Notification and Response Module:

In critical cases (e.g., risk score of at least 60 or failed step-up authentication), this module is used to make automated responses, which can include ending user session or informing security teams to start incident response procedures. Moreover, the notification is also sent to the users and administrators via the identified channels of communication (e.g., email or system alerts).

VI. FUTURE SCOPE

The second step in this project is to enhance the scalability, intelligence and flexibility in order to address the quickly changing cyber threats. Advanced models such as transformers and RNNs can be used to improve the system as more advanced models are developed to detect complex behavior patterns in real time as attacks become more sophisticated. It will also be possible to add blockchain to introduce an auditing log that is tamper-proof and transition to a cloud-native design to allow smooth scaling across hybrid and multi-clouds. Federated learning can also enhance privacy and accuracy through a model by enabling several organizations to simultaneously train models without sharing raw data, enabling active learning, which serves as a response to emerging threats. Adaptive response mechanisms can also be introduced into the system that will automatically isolate attacked devices or accounts to curb attacks.

AccessGuard can also be integrated with the SIEM to provide the single visibility and enable the prompt threat response in the security activities of an organization. With the increase in the usage of edge devices and IoT technologies, the system can be scaled to lightweight, real-time anomaly detection on resource-constrained devices. Such features help to proactively detect threats and act in an intelligent automated manner, which has been investigated recently in the field of behavioral analytics. All in all, AccessGuard can become a holistic, AI-powered Zero Trust model that provides proactive, intelligent, and automated cybersecurity in all digital environments.

VII. FEATURES

The AccessGuard is a proposed system, which has some significant features that can be utilized to improve cybersecurity. It uses a Zero Trust paradigm, according to which there is no trusted user or device by default and all the access requests are analysed constantly Machine learning is used in the system to conduct psychological anomaly detection through which user activity patterns are monitored to detect unusual behaviour in real-time, and the system may challenge, block, or enable a multi-factor authentication .It works based on a Zero Trust paradigm, in accordance to which any trusted user or device is not by default and all the access requests are evaluated at all times. In the system, machine learning is employed to detect any anomaly in psychology whereby user activity pattern would be monitored to identify unusual behavior in real-time, with the system able to challenge, block

or authorize a multi-factor authentication. Additionally, AccessGuard maintains comprehensive security logs and audit trails to support compliance and forensic analysis. System security continues to operate due to its real-time architecture, and proactive threat detection and a flawless user experience are guaranteed.

VIII. APPLICATIONS

A. Enterprise Security

AccessGuard can reinforce corporate security by validating employees, contractors, and third-party vendors through continuous behavioural analysis. [15]This reduces insider threats and protects business-critical systems and confidential data, especially in remote and hybrid work environments. Zero-Trust Security addresses these challenges by enforcing strict access controls, monitoring user behaviour, and leveraging advanced technologies.

B. Financial Institutions

AccessGuard has the ability to detect anomalies in banking and finance sector such as fraudulent transactions, strange logins and others. Irregular fund transfers. The system secures stock trading facility, payment infrastructure and Web-based banking portals with adaptive authentication and user activity monitoring.

C. Healthcare Systems

AccessGuard improves the security of telemedicine applications, electronic health records (EHR), and management systems in the hospital. [16]The constant monitoring of behaviour will stop ransomware and data breaches and will guarantee that it adheres to data protection regulations, including HIPAA and GDPR. Zero-trust not only limits lateral movement within networks but also aligns with compliance requirements such as GDPR and CCPA.

D. Security and Government

The Zero Trust system of AccessGuard is applicable in such sensitive environments as government offices, defense, and intelligence. It prevents cyberattacks and espionage of crucial national information, prevents unauthorized access, and detects abuse of the inside.

IX. CONCLUSION

AccessGuard is an advanced cybersecurity solution, which is based on the Zero Trust model, and it is much more than the old-fashioned defenses based on the perimeter. A combination of real-time behavioral anomaly detection and adaptive access controls can help it counter threats based on stolen credentials, vulnerable devices, or insider misuse. AccessGuard reduces the attack surface by repeatedly examining users and devices and prevents unauthorized access, as well as improves digital resilience overall. The scalability of it enables its use in a wide range of environments including cloud, IoT, government, healthcare, finance, and enterprise systems. AccessGuard offers more powerful protection against future-proof cyberattacks and helps to comply, build stronger trust, and security in a rapidly changing environment.

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